SCH 3U – Unit 2
NAME: ______________________________ DATE:______________________________

Developing an Activity Series Developing an Activity Series

While metals share common properties, they do not all behave the same way in single displacement reactions. For example, zinc displaces many metal cations from their aqueous ionic compounds. Gold, on the other hand, is extremely unreactive. Potassium reacts violently with water. Zinc does not react with water, but it reacts with acids. How can metals be ranked in order of their reactivity?

Question Question

What is the order of reactivity if the metals copper, iron, magnesium, tin and zinc in single displacement reactions?

Materials Materials

- well plates
- six small pieces each of copper, iron, zinc, magnesium, and tin
- dropper bottles of dilute solution of $ZnSO_4$, $MgSO_4$, $SnSO_4$, $FeSO_4$, HCl, and $CuSO_4$.
- test tubes and rack
- wash bottle with distilled water

Procedure

1. Design a procedure to react each metal with each of the given solutions, as well as with water. In your procedure, identify amounts of the materials that you will need to combine and observe. Prepare a data table to record your observations.
2. Have your teacher approve your procedure before you begin.
3. Carry out your procedure. If any of the metals you are given appear dull (rather than shiny) remove this oxide with stainless steel wool.
4. Record any changes in the appearance of each metal or solution. Look for colour changes on the surface of the metal or in the solution. If you have difficulty determining whether a reaction has occurred, repeat the test using a test tube to better observe the reaction. Remember that some reactions are slow so plan to re examine any combinations that to not react immediately.
5. At the end of the investigation, dispose of the solutions in the waste beaker provided. Do not pour anything down the drain.

Procedure

1. Use steel wool to polish all metal pieces until they are shiny.
2. Put 10 drops of each solution (ZnSO4, MgSO4, SnSO4, FeSO4, HCl, CuSO4) and water into separate wells in a well plate.
3. Add one piece of metal to each solution as planned in the data table.
4. Observe for 5-10 minutes. Look for bubbles, color changes, or new solids forming on the metal.
5. Record "Reaction" or "No Reaction" for each combination.
6. Empty the well plate into the designated waste beaker and rinse the plate.

Metal	$MgSO_4$	$ZnSO_4$	$FeSO_4$	$SnSO_4$	$CuSO_4$	HCl	H_2O
Magnesium (Mg)	No reaction	Reaction	Reaction	Reaction	Reaction	Fast bubbles	Slow bubbles
Zinc (Zn)	No reaction	No reaction	Reaction	Reaction	Reaction	Bubbles	No reaction
Iron (Fe)	No reaction	No reaction	No reaction	Reaction	Reaction	Slow bubbles	No reaction
Tin (Sn)	No reaction	No reaction	No reaction	No reaction	Reaction	Very slow	No reaction
Copper (Cu)	No reaction	No reaction	No reaction	No reaction	No reaction	No reaction	No reaction

Analysis Analysis

1. For any reactions that occurred, write the corresponding balanced single displacement reaction.

Magnesium (Mg) Reactions:

- $Mg + ZnSO_4 \rightarrow MgSO_4 + Zn$
- $Mg + FeSO_4 \rightarrow MgSO_4 + Fe$
- $Mg + SnSO_4 \rightarrow MgSO_4 + Sn$
- $Mg + CuSO_4 \rightarrow MgSO_4 + Cu$
- $Mg + 2HCl \rightarrow MgCl_2 + H_2$

Zinc (Zn) Reactions:

- $Zn + FeSO_4 \rightarrow ZnSO_4 + Fe$
- $Zn + SnSO_4 \rightarrow ZnSO_4 + Sn$
- $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

- $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Iron (Fe) Reactions:

- $Fe + SnSO_4 \rightarrow FeSO_4 + Sn$
- $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
- $Fe + 2HCl \rightarrow FeCl_2 + H_2$

Tin (Sn) Reactions:

- $Sn + CuSO_4 \rightarrow SnSO_4 + Cu$
- $Sn + 2HCl \rightarrow SnCl_2 + H_2$

2. Lithium metal reacts with water.
    a) Is lithium more or less reactive than magnesium?
        - Lithium is more reactive than magnesium because lithium reacts immediately with room temperature water, whereas magnesium typically requires hot water for a significant reaction.
    b) Write a balanced equation to represent the reaction of lithium with water.
        - $2Li + 2H_2O \rightarrow 2LiOH + H_2$
3. Given that lithium reacts with water and magnesium does not, do you expect lithium to react with hydrochloric acid? Explain your answer.
    - lithium will react with HCl. Since lithium is reactive enough to break the bonds in water to release hydrogen gas, it will react even more with a strong acid like HCl, which has a much higher concentration of hydrogen ions.
4. Why do you think that solutions containing only sulfate ions were used?
    - Sulfate ions were used to keep the anion consistent. Since every solution used the same sulfate ion, any reaction (or lack of reaction) can be explained entirely on the metal's reactivity rather than the solution's composition.

Conclusion

1. Write the activity series corresponding to your observations. Include hydrogen as an ion (H^+).
    - $Mg > Zn > Fe > Sn > H+ > Cu$
2. How do you think an activity series for metal would help you predict whether or not a single displacement reaction will occur? Use examples to help you explain your answer.

    - The activity series is a ranking system. A metal can only displace another metal if it is higher up on the list.
        - Ex: If you put Zinc into a Copper solution, a reaction occurs because Zinc is higher than Copper.
        - Ex 2: If you put Copper into a Zinc solution, no reaction occurs because Copper is lower and does not have enough energy to displace the Zinc.